INCH-POUND

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21 September 2011

# PERFORMANCE SPECIFICATION

# LIFE RAFT, INFLATABLE, SINGLE-PLACE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for an inflatable one-man, single-place life raft.

1.2 Classification. The life rafts are of the following types as specified (see 6.2).

1.2.1 Types. The types of single-place life rafts are as follows:

Type I - Navy Aircraft
Type II - Army Helicopter
Type III - Navy Aircrew Common Ejection Seat (NACES)
Type IV - NACES Emergency Enhanced Oxygen System (NACES EEOS)

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division Lakehurst, Code 4.1.2.2, Route 547, Mail Stop 120-3, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-A-3339 | - | Anchors, Sea, Life Raft. (Inactive for New Design) |
| MIL-DTL-32075 | - | Label: For Clothing, Equipage, and Tentage (General Purpose) |
| MIL-DTL-81722 | - | Valve, Inflation, Life Raft, (FLU-6/P) |
| MIL-C-83489 | - | Cloth, Coated, Nylon, Polyurethane Coated |

DEPARTMENT OF DEFENSE STANDARD

| | | |
|---|---|---|
| MIL-STD-810 | - | Environmental Engineering Considerations and Laboratory Tests |

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND

| | | |
|---|---|---|
| NAVAIR 13-1-6.3-1 | - | Aviation Crew Systems Manual Seat Survival Kits |
| NAVAIR 13-1-6.3-2 | - | Aviation Crew Systems Manual Seat Survival Kits |

(Copies of these documents can be obtained from the contracting activity or as directed by the contracting officer.)



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2.3 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes. (DoD adopted)

(Copies of this document are available from www.asq.org.)

SAE INTERNATIONAL

SAE AMS-STD-595/12197 – Orange Gloss
SAE AMS-STD-595/28915 – Fluorescent Orange
SAE AMS-STD-595/27040 – Filing Cabinet Black
SAE AMS-STD-595/35042 – Blue, Flat or Lusterless

(Copies of these documents are available online at http://sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Design. The individual life raft shall be easily inflatable with a canopy, anchor, and pockets and shall accommodate military aircrew personnel. The life raft shall be manufactured and designed to incorporate a cost effective life cycle with respect to maintenance, shelf life, repack, and/or replacement. The life raft shall meet all the requirements specified herein.

3.2 Qualification. The life raft furnished under this specification shall be a product that is authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 First article. When required (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.4 Materials. The material of the raft main buoyancy tube shall conform to MIL-C-83489, type I. The latest technology materials and processes shall be utilized. The use of ozone depleting substances (ODS) shall be limited. Unless otherwise specified, the materials and



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components, except for the metallic parts, used in the construction of the life raft shall not have been manufactured more than 18 months prior to the delivery of the life raft (see 6.2).

3.5 Inflation system. An inflation system meeting the performance requirements of MIL-DTL-81722 or equivalent shall be utilized. The inflation system shall not leak during the manufacturer's established raft repack/recertification cycle.

3.5.1 Secondary inflation. A means readily accessible to the life raft occupant shall be provided to complete inflation (top-off) and provide emergency inflation to maintain operating pressure. The means shall be properly located, reinforced, and protected against any possible leakage or damage that may occur due to reduced volume packing.

3.6 Canopy. The life raft design shall incorporate an inflatable canopy system to provide aircrew protection from sea spray and extreme weather conditions. The canopy shall provide adequate headroom based on the anthropometric data in table II and shall be made weathertight. A clear visor or panel shall be incorporated to allow the aircrewman vision when the canopy is made weathertight. Raft buoyancy shall not be affected by the integrity of the canopy system. The canopy shall be in a stowed position (secured) during inflation and shall be manually inflated by the aircrewman after boarding.

3.6.1 Floor. The floor of the life raft shall be insulated to provide cold protection to the occupant. This may be achieved with an inflatable area that covers approximately 90 percent of the usable floor area. Raft buoyancy shall not be affected by the integrity of the floor. The floor shall be orally inflated by the aircrewman after boarding.

3.7 Sea anchor. A sea anchor shall be incorporated into the raft design to maintain the raft in a sea environment. The sea anchor shall conform to MIL-A-3339, type I or equivalent. The sea anchor shall maintain the raft on a constant heading relative to the wind and have the ability to reduce drift in 17- to 27-knot winds. The sea anchor shall be stowed (held in place) for the aircrewman to deploy after boarding.

3.8 Stowage pockets. Stowage pockets shall be incorporated into the life raft to hold the strobe light, lanyard, and the radio beacon. Stowage pocket dimensions (L x W x D in inches) shall be in accordance with table I.

3.8.1 Strobe light pocket. The strobe light pocket shall be located on the outside of the raft for signaling purposes. The strobe light pocket on the exterior canopy shall be a good match to SAE AMS-STD-595/35042 (lusterless blue) (see 3.13).



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3.8.2 Radio beacon pocket. The radio beacon pocket, located on the interior tube, shall be a good match to SAE AMS-STD-595/12197 (orange gloss) or SAE AMS-STD-595/28915 fluorescent orange (see 3.13).

3.8.3 Lanyard pocket.The lanyard pocket, located on the exterior tube near the inflation system, shall be a good match to SAE AMS-STD-595/35042 (lusterless blue) (see 3.13).

TABLE I. Stowage pockets.

| Pocket type | Type I, II, III | Type IV | Tolerance |
|---|---|---|---|
| Strobe light | 5 x 3.5 x 1 | 3.25 x 2 x 1 | ± ¼ |
| Radio beacon | 8 x 4 x 2 | 10.8 x 5 x 2 | ± ¼ |
| Lanyard | N/A | 3.8 x 2.5 x 0.5 | ± ¼ |
| Dimensions (L x W x D) are in inches. | | | |

3.9 Equipment container attachment. A means shall be provided to secure the survival equipment container to the inside of the raft.

3.10 Capsize resistance. The life raft shall have ballast bags or other means to provide stability and capsize resisitance.

3.11 Righting aid. A means shall be provided to right the life raft if it inflates in an inverted position. The means provided for righting the raft shall be such that it can be activated by one person in the water.

3.12 Life raft equipment. All lines shall be stowed and secured to prevent entanglement during launching/inflation of the life raft.

3.13 Color. The color of the life raft exterior surfaces, including the canopy surface, shall be blue/black in nature and shall be in accordance with SAE AMS-STD-595/35042 (lusterless blue) or SAE AMS-STD-595/27040 (black). High-visibility orange in accordance with SAE AMS-STD-595/12197 or /28915 shall be used on the inside of the canopy for rescue operations.

3.14 Capacity. The life raft shall have adequate capacity for an aircrewman wearing standard aviation life support equipment including a flight suit, survival vest, inflated life preserver, and flight boots. Anthropometric characteristics are presented in table II.



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TABLE II. Anthropometric characteristics.

| Measurements (in inches) | Case 1 | Case 2 | Case 3 | Case 4 | Case 5 | Case 6 | Case 7 |
|---|---|---|---|---|---|---|---|
| | Overall small | Short sitting heights with long limbs | | Large sitting height with short limbs | Overall large | Longest limbs | Shortest torso |
| Thumb tip reach | 27.0 | 27.6 | 33.9 | 29.7 | 35.6 | 36.0 | 26.1 |
| Buttock-knee length | 21.3 | 21.3 | 26.5 | 22.7 | 27.4 | 27.9 | 20.8 |
| Knee-height sitting | 18.7 | 19.1 | 23.3 | 20.6 | 24.7 | 24.8 | 18.1 |
| Sitting height | 32.8 | 35.5 | 34.9 | 38.5 | 40.0 | 38.0 | 31.0 |
| Eye-height sitting | 28.0 | 30.7 | 30.2 | 33.4 | 35.0 | 32.9 | 26.8 |
| Shoulder-height sitting | 20.6 | 22.7 | 22.6 | 25.2 | 26.9 | 25.0 | 19.5 |
| Shoulder breadth range | 14.7-18.1 | 16.4-20.6 | 16.2-21.2 | 16.8-21.7 | 16.9-22.6 | 16.8-22.5 | 14.2-18.0 |
| Chest depth range | 7.4-10.9 | 6.9-10.6 | 7.2-11.3 | 7.1-11.0 | 7.3-12.1 | 7.4-12.2 | 7.2-10.2 |
| Thigh circumference range | 18.5-25.0 | 17.1-25.0 | 20.2-27.6 | 17.6-26.3 | 18.6-29.2 | 19.1-29.7 | 17.8-25.2 |
| Nude body weight range | 103 pounds to 245 pounds | | | | | | |

3.15 Buoyancy. The life raft shall provide a freeboard of not less than 2 inches at its lowest point immediately after boarding with a fully equipped aircrewman in accordance with table II.

3.16 Weight. The weight of the packaged life raft including the inflation assembly shall be not greater than 10 pounds.

3.17 Stowage.

3.17.1 Type I - Navy aircraft. The Type I life raft, when folded, shall be packed in Navy seat kits and a Navy helicopter backpack. Available volume for the life raft is in accordance with applicable NAVAIR seat kit and helicopter backpack manuals.



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3.17.2 Type II – Army helicopter. The Type II life raft shall be sealed in reduced volume packaging. The seal shall provide protection and preservation for the raft in an aircraft environment. The life raft shall be sealed in such a way as to guarantee verification of airtight packaging by visual methods.

The Type II life raft shall be packed in the helicopter aircrewman's Air Warrior Over-Water Gear Carrier (OWGC). The inflation cord attachment shall be located such that the lanyard exits the pack at the upper left corner to allow activation over the left shoulder of the aircrewman. The packed configuration shall be in accordance with figure 1.

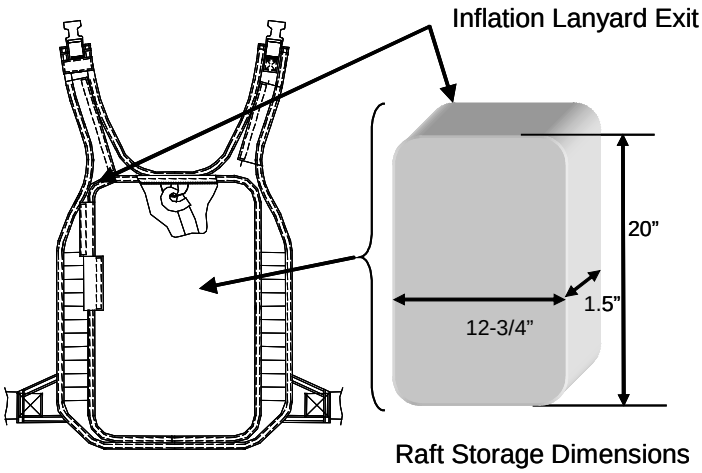



FIGURE 1. Type II - Army helicopter.

3.17.3 Type III – NACES ejection seats. The Type III life rafts, when folded, shall be packed in Navy NACES ejection seats only. Available volume for the life raft is in accordance with applicable NAVAIR seat kit manual chapters.

3.17.4 Type IV – NACES EEOS. The Type IV life rafts, when folded, shall be packed in Navy NACES ejection seats only. Available volume for the life raft is in accordance with applicable NAVAIR seat kit manual chapters.

3.18 Actuation. All life raft types shall actuate manually by pulling an inflation lanyard. The pull force required for lanyard actuation shall be not greater than 30 lb., when tested in accordance with 4.10. Type I, III, and IV life rafts shall also actuate automatically by gravity drop on kit actuation when jump tested in accordance with 4.11.

3.19 Raft pressure.



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3.19.1 Pressure. The pressure in the raft flotation tube shall be not less than 95 percent of 2 times the manufacturer's specified operating pressure when tested in accordance with 4.12.1.

3.19.2 Leakage. The pressure in the raft flotation tube shall be not less than 80 percent of the manufacturer's specified operating pressure when tested in accordance with 4.12.2.

3.20 Function/operation. The life raft shall inflate at the required pull force to its design shape in not more than 60 seconds when tested as specified in 4.13. During the inflation, the life raft shall be observed for impediment or blockage of the flow of gas, loss of gas to the ambient air, or restriction by any component or accessory. There shall be no evidence of structural or material failure in any respect. All the sealed areas, seams, seam tapes, and attachments shall remain intact and shall show no indication of separation. The floor shall not be distorted and there shall be no difference in the rise between the sides of the raft. The erected canopy shall withstand 35-knot winds and 52-knot gusts in open water. The life raft shall demonstrate seaworthiness and stability in an open sea condition of 17 to 27-knot winds and waves of 6 to 10 feet. The raft shall be easily boarded by a subject wearing standard aviation life support equipment including a flight suit, survival vest, inflated life preserver, and flight boots. See table II for anthropometric data.

3.21 Environmental.

3.21.1 Temperature extremes.

3.21.1.1 Low temperature. The life raft shall inflate into a boardable shape within 5 minutes at 0 °F (-18 °C), when tested as specified in 4.14.1.1. There shall be no hindrance to the flow of gas or restriction by any component or accessory during inflation. All the seams, seam tapes, sealed areas and attachments shall remain perfectly intact and shall show no evidence of separation. There shall be no evidence of construction or material failure in any respect.

3.21.1.2 High temperature. The life raft shall inflate to design shape within 60 seconds at 160 °F (71°C) when tested as specified in 4.14.1.2. Inflation shall be without any hindrance to the flow of gas or restriction by any component or accessory. All the seams, seam tapes, sealed areas and attachments shall remain perfectly intact and shall show no evidence of separation. There shall be no evidence of construction or material failure in any respect.

3.21.2 Low pressure (altitude). The packed life raft shall withstand a low pressure environment and rapid pressure changes when tested as specified in 4.14.2. The life raft shall show no leakage of gases and no rupture or failure of sealed packaging.

3.21.3 Vibration. The packed life raft shall resist the effects of aircraft vibration when tested as specified in 4.14.3. The life raft shall show no leakage of gases and no rupture or failure of sealed packaging.



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3.21.4 Accelerated aging/life cycling. The packed life raft shall withstand the effects of accelerated aging/life cycling when tested as specified in 4.14.4. The life raft shall inflate to design shape within 60 seconds when actuated as specified in 4.14.4. Inflation shall be without any hindrance to the flow of gas or restriction by any component or accessory. All the seams, seam tapes, sealed areas and attachments shall remain perfectly intact and shall show no evidence of separation. There shall be no evidence of construction or material failure in any respect. Immediately following this test, the raft shall be visually examined as specified in 4.7, and pressure/leakage tested as specified in 4.12.

3.22 Markings.

3.22.1 Operational marking. The life raft shall be marked in contrasting washproof ink, which is not detrimental to the fabric, to denote the use and location of the inflation systems (including instructions on manual actuation), raft equipment, boarding aids and righting aids. The letters used for such markings shall be not less than 1 inch high except that details and miscellaneous instructions may be of smaller lettering. Applicable markings shall take into account aircrewmen boarding or righting the raft from the water, and shall be in accordance with MIL-DTL-32075.

3.22.2 Raft identification. Each life raft shall be identified by name of manufacturer, contract number, date of manufacturer, and serial number. The serial number shall be assigned by the manufacturer and shall be by a block of consecutive numbers to cover the entire acquisition quantity. All markings shall be legible, durable, permanent, and thoroughly dry prior to packing and packaging.

3.22.3 Bar code. Each raft shall be individually barcoded with a Type VII (bag), Type VIII (tag), or Type IX (heat sealed) label/tag, as specified in MIL-DTL-32075, clearly legible and readable by scanner. The bar code element shall be a 13-digit national stock number (NSN). There shall be a 12-digit Universal Product Code (UPC) assigned for all NSNs by the Government. The initial “UPC” shall appear beneath the code. The bar codes for the NSN and the UPC shall be a medium to high density and shall be located so that they are completely visible on the item when it is folded and/or packaged as specified. The label shall not cause damage to the raft.



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## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. First article inspection (see 4.3).
c. Conformance inspection (see 4.4).

4.2 Qualification inspection. The qualification inspection of the life rafts shall consist of all the examinations and tests specified in table III.

4.2.1 Qualification samples. The qualification samples shall consist of three of each type to be qualified. Samples shall be forwarded as specified in the correspondence authorizing submission of samples for qualification testing (see table III and 6.2). Additionally, each sample shall be plainly identified by securely attached tags containing the following information:

Qualification inspection sample
Raft type
Manufacturer's designation
Manufacturer's name and CAGE
Serial number

TABLE III. Qualification examinations and tests.

| Inspection | Requirement | Test method |
|---|---|---|
| Weight | 3.16 | 4.8 |
| Stowage | 3.17 | 4.9 |
| Actuation pull force test | 3.18 | 4.10 |
| Jump test | 3.18 | 4.11 |
| Pressure | 3.19.1 | 4.12 |
| Leakage | 3.19.2 | 4.12.2 |
| Operational testing | 3.14, 3.15, 3.20 | 4.13 |
| Visual examination | 3.4-3.13, 3.22 | 4.7 |
| Temperature extremes test | 3.21.1 | 4.14.1 |
| Low pressure (altitude ) test | 3.21.2 | 4.14.2 |
| Vibration | 3.21.3 | 4.14.3 |
| Accelerated aging/life cycle | 3.21.4 | 4.14.4 |

4.3 First article inspection. First article inspection, when required, shall be accomplished as specified herein. The first article inspection shall consist of the tests and examinations specified in table IV (see 6.2).



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TABLE IV. First article examinations and tests.

| Inspection | Requirement | Test Method |
|---|---|---|
| Weight | 3.16 | 4.8 |
| Stowage | 3.17 | 4.9 |
| Actuation pull force test | 3.18 | 4.10 |
| Pressure | 3.19.1 | 4.12.1 |
| Leakage | 3.19.2 | 4.12.2 |
| Visual examination | 3.4-3.13, 3.22 | 4.7 |

4.4 Conformance inspection. The conformance inspection shall consist of the tests and examinations specified in table V (see 6.2).

4.5 Inspection methods.

4.5.1 In-process inspection. Examination shall be made at any point or during any phase of manufacturing process to determine whether construction details that cannot be examined in the finished product are in accordance with requirements specified in Section 3. Materials and components that can be classified as a defect shall be removed from production.

4.5.2 Conformance inspection samples. Unless otherwise specified, sampling for inspections shall be performed in accordance with ANSI/ASQ-Z1.4. The sample unit shall be one raft. The lot size shall be expressed in units of rafts.

TABLE V. Conformance examinations and tests.

| Inspection | Requirement | Test method |
|---|---|---|
| Weight | 3.16 | 4.8 |
| Stowage | 3.17 | 4.9 |
| Actuation pull force test | 3.18 | 4.10 |
| Pressure | 3.19.1 | 4.12 |
| Leakage | 3.19.2 | 4.12.2 |
| Visual Examination | 3.4-3.13, 3.22 | 4.7 |



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4.6 Inspection conditions for leakage testing.

4.6.1 Atmospheric conditions. Unless otherwise specified in the contract, all pressure/leakage tests required by this specification shall be conducted at a barometric pressure of 28 to 32 inches of mercury and at a temperature of 77 ±18 °F (25 ±10 °C). If the final values of the ambient temperature or barometric pressure at the end of the four-hour leakage inspection are different from the initial values recorded at the start of the inspection, the following corrections should be made to the final pressure readings in psig (see 4.6.2 and 4.6.3).

4.6.2 Temperature correction. For each degree Fahrenheit rise in temperature, 0.031 psig shall be subtracted from the final pressure reading. For each degree Fahrenheit drop in temperature, 0.031 psig shall be added to the final pressure reading. The corresponding correction per degree Celsius is 0.056 psig.

4.6.3 Barometric pressure correction. For each 0.1 inch of mercury rise in barometric pressure, 0.049 psig shall be added to the final temperature-corrected pressure reading. For each 0.1 inch of mercury drop in barometric pressure, 0.049 psig shall be subtracted from the final temperature-corrected pressure reading.

4.7 Visual examination. The visual examination shall be performed to verify conformance to the requirements specified in section 3. Table VI classifies and enumerates the defects.

4.8 Weight. The weight of the life raft shall meet the requirements of 3.16, when weighed on a calibrated scale.

4.9 Packaging for stowage. The raft in its packed configuration for stowage shall be demonstrated to meet the requirements of 3.17.



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TABLE VI. Classification of defects for visual examination of the life rafts.

| Critical | Minor |
|---|---|
| 1. Any hole, cut, tear, patch, or burn | 201. Any spot or stain. 1/ |
| 2. Any fabric damaged, bruised, abraded, containing imperfections, or otherwise defective. 1/ | 202. Any cut edge of uncoated nonfibrous material containing sharp edges. |
| 3. Excessive bias in fabric resulting in twisting of primary buoyancy tube. 2/ | 203. Any metal component (e.g., slide fastener) improperly installed or inoperable. |
| 4. Any stitching in the inflatable section of the life raft. | 204. Stitching damaged or defective, irregular or run-off stitching. |
| 5. Attachment of the floor or canopy to the inflatable tube not reinforced or not attached. | 205. Any pile or hook tape damaged or defective. |
| 6. Inflation assembly or manifold stem assembly not in accordance with requirements, or inoperable. | 206. Topping-off inflation assembly or components not as specified; topping-off valve not locked in the closed position. |
| 7. Any pillowing of the sealed packed raft by inflation system leakage or loss of integrity of sealed vacuum packaging. | 207. Any required markings, illegible, incomplete, incorrect, not readable by scanner, improperly located or not visible on fold or packaged end item. |
| 8. An inflatable section seam separation or construction that does not meet the specified minimum requirements. | 208. Color of any component not as specified. |
| 9. Any channels or voids in any seam on the inflatable section. | 209. Accessory or other container not closed. |
| 10. Topping-off valve damaged or in open position. | 210. Life raft improperly folded. |
| 11. Any metal component improperly finished or containing nicks, burrs, dents, sharp edges, or rough surfaces. 1/ | 211. Any wrinkles, channels, or voids in any seam, patch or attachment on the inflatable section. 1/ |
| 12. Any component, component part, or required operation omitted; or any operation improperly performed, not herein classified. 1/ | 212. Cement on the cloth surfaces around patches or attachments. |
| 13. Any component not as specified, or any defect of a component or assembly not herein classified. 1/ | 213. Any clot or mass of adhesive. |
| | 214. Overlap of any seam tape in excess of requirements. |
| | 215. Any seam tape, patch or attachment separating. 1/ |

1/ The defect is classified critical when it affects serviceability or function; otherwise, it is classified minor.

2/ If excessive twisting of the primary buoyancy tube is evident, the life raft shall be examined as follows: With the primary buoyancy tube inflated to its specified operating pressure and



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with the floor and the canopy (if applicable) deflated, the life raft shall be placed on a horizontal surface. A 3-inch block shall be placed under the life raft at two diagonally opposite locations. A gap between the bottom of the primary buoyancy tube and the 3-inch high block is evidence of excessive bias in the fabric and shall be classified as a critical failure.

4.10 Actuation pull force. Actuation of the life raft shall be tested both for manual and automatic actuation (kit release). The actuation shall meet the requirements of 3.18.

4.11 Jump test. Type I, III, and IV life rafts shall be evaluated for automatic actuation (kit release) by performing a jump test. Each applicable packed configuration shall meet the requirements of 3.18. A test subject shall be fully dressed in aviation life support gear with a seat kit attached. The test subject shall simulate parachute descent on a slide tower and perform test kit release. The slide tower must be approved by the qualifying activity.

4.12 Raft pressure.

4.12.1 Pressure. The completely deflated life raft shall be placed in a horizontal position on the floor or table of the inspection area. All the pressure readings shall be taken with the raft in this position. The flotation tube shall be inflated, with air, to 2 times the manufacturer's specified operating pressure. The air supply shall be securely shut off and after a minimum of 10 minutes, the pressure in the raft flotation tube shall be checked and readjusted, if necessary, to the original pressure. At the end of a minimum of 10 minutes after the readjustment period, the pressure in the tube shall be not less than 95 percent of 2 times the manufacturer's specified operating pressure. While inflated, the raft shall be visually examined (see 4.7) and meet the requirements of 3.19.1. Upon completion of the testing and examination, the raft flotation tube shall be completely deflated and the raft shall then be subjected to the leakage inspection.

4.12.2 Leakage. The completely deflated life raft shall be placed in a horizontal position on the floor or table of an inspection area. The temperature and pressure readings shall be taken with the raft in this position and in the vicinity of the raft. The inflatable tube shall be inflated, with air, to the manufacturer's specified operating pressure. The air supply shall be securely shut off and after a minimum of 15 minutes, the pressure in the tube shall be checked and readjusted, if necessary. At the end of a minimum of 4 hours, after the readjustment period, the pressure shall be measured and corrected for any change in the temperature or barometric pressure (see 4.5). The corrected pressure in the raft flotation tube shall be not less than 80 percent of the operating pressure. The raft shall meet the requirements of 3.19.2.

4.13 Operational tests. In either a controlled pool, fresh water, or at sea, the life raft operation shall be demonstrated as follows: Capacity and buoyancy shall be verified for conformance with the requirements of 3.14 and 3.15. Function/operation of the life raft shall be



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evaluated for its conformance to 3.20. The life raft's components and features shall meet the requirements specified in section 3.

4.14 Environmental tests.

4.14.1 Temperature extremes test.

4.14.1.1 Low temperature test. The life raft shall be conditioned at 0 ±2 °F (-18 ±1 °C) for not less than 48 hours. The raft shall then be removed to the inspection area and placed on a table or on the floor. The inflation equipment shall be immediately actuated. The life raft shall meet the requirements of 3.21.1.1.

4.14.1.2 High temperature test. The test procedures set forth in 4.14.1.1 shall be repeated, except the raft shall be conditioned at 160 ±2 °F (71 ±1 °C) instead of 0 °F, for not less than 48 hours. The raft shall meet the requirements of 3.21.1.2.

4.14.2 Low pressure (altitude) test. The packed life raft shall be tested in a low pressure environment in accordance with MIL-STD-810, Method 500, Procedure III. At the completion of the test, the raft shall be removed from the low pressure environment. The raft shall meet the requirements of 3.21.2.

4.14.3 Vibration test. The packed life raft shall be tested for its ability to withstand aircraft vibration in accordance with MIL-STD-810, Method 514. The vibration levels shall be tailored to represent specific aircraft in accordance with the applicable specification sheets. At the completion of the vibration test, the life raft shall meet the requirements of 3.21.3.

4.14.4 Accelerated aging/life cycling test. The accelerated aging/life cycle test shall include altitude cycling between 0 and 12,000 ft. at a constant accelerated temperature between 158 °F to 170 °F (70 °C to 77 °C). Based on 200 flights per year, the performance of 1,000 altitude cycles during the temperature soak should provide a five-year life. The number of cycles shall be contingent upon the manufacturer's proposed life cycle. An altitude rate of change of 2,000 ft. per minute and a one-minute dwell at 12,000 ft. altitude shall be performed. The raft shall then be removed to the inspection area and placed on a table or on the floor. The inflation equipment shall be immediately actuated. The life raft shall meet the requirements of 3.21.4.



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The life rafts are intended for use as emergency equipment by military aircrews forced down at sea. Commercial life rafts do not meet the stringent protection and packaging requirements necessary for military aircraft use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of raft (see 1.2.1).

c. Whether first article inspection is required (see 4.3).

d. Certificate of compliance for the age of the materials and components (see 3.4).

e. Qualification samples (see 4.2.1).

f. National stock number (NSN).

g. Color required.

h. Packed raft, shelf life.

i. Applicable manuals and specification sheets.

j. When physical testing is required for lot acceptance.



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k. Name and address of the conformance verification inspection facility (see 4.4); and the name and address of the Government activity responsible for conducting the conformance verification inspection program (see 4.4).

l. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-81542 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Air Warfare Center Aircraft Division, Code 4.6.7.4, 48110 Shaw Road, Bldg. 2187, Patuxent River, MD 20670. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Canopy
Seals
Survival

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
DLA - CT

Preparing activity:
Navy - AS
Project 4220-2019-001

Review activity:
Navy - NU

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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